



P20

**The Diversity Dividend and
Homogeneous Saturation: An
Information-Theoretic Account
of Why Independent Information
Sources, Not Headcount, Are the
Quality Variable in Multi-Agent
Systems**



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Working Title (EN): The Diversity Dividend and Homogeneous Saturation: An Information-Theoretic Account of Why Independent Information Sources, Not Headcount, Are the Quality Variable in Multi-Agent Systems

Volume P20 • external dialogue paper to P9 "The Quality Saturation Law of Agent Legions and the Failure Boundary of Homogeneous Headcount Expansion" • sister paper to P22 "The Memory Wall Is Not the Coordination Wall" • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: piling ten thousand "clone Agents" of the same model, same prompt set, same context together does not change quality—because they see the same world; what truly sets a multi-Agent system's ceiling is not the Agent **count** $\$N\$$ but the number of independent **information sources** $\$D\$$. OASIS observed with million-scale heterogeneous social individuals that "larger is more diverse," UDOS P9 measured with a cpu-proto experiment that "homogeneous copies saturate no matter how many added," TUMIX proved with heterogeneous tool strategies that diversity brings measurable gains—three independent evidence lines point to the same information-theoretic conclusion.

Evidence caliber (stated once for the whole paper): this is a theoretical/synthesis paper, containing no new UDOS measured numbers; cited UDOS conclusions all traced back to v7.5.0 papers and their reports7/*.json three-level labels (verified / cpu-proto / cpu-proxy / unverified). External literature (OASIS, TUMIX, Krogh & Vedelsby, Kaplan, Chinchilla, NSA, etc.) existence verified 2026-09-21 per EVIDENCE_LEDGER_v76.md; OASIS's "larger group $\rightarrow$ more diverse" and TUMIX's "+3.55%" both paper-report caliber, not independently rechecked. This paper writes no external number as independent evidence for its argument, only as corroboration that "industry phenomena align with UDOS inferences."

Abstract

The popular narrative of multi-Agent systems is "headcount equals strength": as long as the parallel sub-Agent count $\$N\$$ rises, accuracy, robustness, coverage rise with it. This paper argues this narrative conflates two different quantities—**scale** (headcount $\$N\$$) and **independent information source count** (diversity $\$D\$$). The argument in four steps. First, the ensemble learning classic saturation law $\$mse(k)=a+b/k\$$ shows: the ensemble error decline $\$b/k\$$ is set by member error independence; when members are homogeneous (error correlation $\$r\approx 1\$$), $\$b\rightarrow 0\$$, adding more members pins quality at the irreducible floor $\$a\$$; UDOS P9's cpu-proto measurement reproduces this shape ($\$a=0.009252\$$, $\$b=-4.23\times 10^{-5}\$$, flattening after $\$k>2\$$). Second, OASIS (arXiv:2411.11581, NeurIPS 2024) in million-scale social simulation observed "larger Agent groups bring stronger group dynamics and more diverse opinions," but that system's Agents are **heterogeneous individuals** (different user profiles, different behavior rules), contrasting with P9's **homogeneous copies** (same model, same prompt, $\text{corr}\approx 0.9999$)—the two evidence lines together make the point: scale itself does not set diversity, inter-individual information independence does. Third, TUMIX (arXiv:2510.01279) on Gemini-2.5-Pro with "parallel multi-Agents, each using a different tool strategy" heterogeneous ensemble averages up to +3.55% gains, proving heterogenization an operable quality lever. Fourth, the engineering corollary: before expanding headcount ask "does the new Agent carry information existing Agents don't see," not "can I spin up one more instance"; Warp-Cortex-class singleton weight sharing architectures (P22) naturally carry no independent information, their value in memory not quality. This paper gives three falsifiable conditions and two hard predictions.

Keywords: multi-Agent systems; ensemble saturation law; diversity dividend; independent information sources; homogeneous headcount expansion; bias-variance-ambiguity decomposition; falsifiability

Structured Abstract (background problem $\rightarrow$ argument $\rightarrow$ evidence $\rightarrow$ contribution)

- **Background problem:** multi-Agent engineering long sprinted down the "add people" path, but what does adding people buy? Quality, throughput, or just the bill? Existing literature either reports only end-to-end success (hiding whether members are truly independent) or only throughput curves (unrelated to quality).

- **Core argument:** the first-principles variable of multi-Agent quality is **independent information source count \$D\$**, not total Agent count \$N\$; homogeneous copies enlarge \$N\$ without increasing \$D\$, the quality curve flattening by \$1/k\$; heterogeneous individuals enlarge \$D\$, quality having room to keep declining.
 - **Evidence lines:** (1) UDOS P9 cpu-proto: homogeneous copies $mse(k)=a+b/k$, flattening after $k>2$, equal-weight bad source drags about $23\times$; (2) OASIS million-scale heterogeneous social simulation "larger is more diverse" observation (paper report); (3) TUMIX heterogeneous tool strategy ensemble $+3.55\%$ (paper report); (4) theoretical ancestor Krogh & Vedelsby 1995 bias-variance-ambiguity decomposition.
 - **Contribution:** separately operationalizing "scale \$N\$" and "independent information sources \$D\$" in one framework; using OASIS (heterogeneous $\rightarrow$ diverse) and P9 (homogeneous $\rightarrow$ saturated) contrast to expose the scale narrative conflation; giving the "before expanding ask information independence" engineering criterion; two hard predictions.
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1 Introduction: Translating "Add People" into "Add What"

"We already run a few hundred sub-Agents in parallel"—this sentence in 2026's multi-Agent engineering context almost equals "we are strong." But what does it measure? More independent judgments, or more repeated votes of the same judgment?

This is not word-splitting. Distributed decision has a repeatedly verified saturation law: averaging multiple independent errors presses variance down, averaging multiple **correlated** errors does not. The former is the whole reason ensemble learning works, the latter the whole reason "adding people has no effect." The problem is, in LLM Agent legion engineering practice, "adding one new Agent" is often **making one more same-model call, feeding the same context**—it and the original Agent almost see the same world. Such "adding people" increases \$N\$ (headcount), not \$D\$ (independent information source count).

UDOS P9 already measured this curve in a controlled prototype: equal-weight including one bad expert can worsen ensemble MSE from about 0.0092 to 0.2131 (about $23\times$), while after $k>2$, 15 more homogeneous copies, MSE flattens around the saturation floor $a=0.009252$ (cpu-proto, single-seed pilot). This paper puts this cpu-proto curve back into 2026's external evidence landscape: OASIS's million-scale social simulation, TUMIX's heterogeneous tool ensemble, from two opposite directions corroborating "independent information sources are the quality variable."

The whole paper deliberately distinguishes three statement types: **UDOS engineering evidence** (to P9, with evidence grade), **external paper reports** (OASIS/TUMIX, marked "paper report • not independently rechecked"), **theoretical inferences** (marked as inferences with falsification conditions in Section 8).

2 Core Argument: Two Conflated Quantities

2.1 Operationalizing \$N\$ and \$D\$

- **\$N\$ (scale / headcount):** total Agent instances simultaneously in the system. It sets compute cost, communication fan-in, memory footprint—i.e. "how many resources to maintain this legion."

- **D (independent information source count)**: the number of mutually non-redundant information channels these Agents rely on. It sets how much error decline the ensemble can squeeze from diversity—i.e. "how far this legion's quality can go."

N and D need not grow together. Copying the same model instance into 1000 containers, $N=1000$, D may still be about 1 (everyone sees the same parameters, the same context); putting 10 Agents of different models, each using different tools, fed different data sources, $N=10$, D may approach 10.

2.2 Why D Is the Quality Variable

Following P9's derivation: the variance of k members' weighted average prediction, when member errors are pairwise independent, zero-mean, variance σ^2 , drops to σ^2/k ; plus the irreducible bias a all members share, total MSE takes the form

$mse(k) = a + \frac{b}{k}$, $\quad a$ \text{the irreducible floor}, $\frac{b}{k}$ \text{the variance decline bought by independence}.

The key inference hidden in b : b proportional to member error independence. When member error correlation $\rho \rightarrow 1$ (homogenization), the variance term no longer declines with k , $b \rightarrow 0$, hence $mse(k) \rightarrow a \quad (\rho \rightarrow 1)$.

That is, **homogeneous copies' quality curve is not "declining slower," but "not declining at all."** Krogh and Vedelsby (NeurIPS 1995)'s bias-variance-ambiguity decomposition wrote this in classic form: ensemble generalization error = average single-model error – inter-member ambiguity; larger ambiguity, better ensemble. b is ambiguity's incarnation in $1/k$ form.

2.3 Distinguishing from Kaplan Scaling Shape

Must distinguish this $1/k$ saturation law from Kaplan's model-size power law ($L(N) = (N_c/N)^\alpha$), else easy to misread. Kaplan describes **switching to a larger model** with continuous, approximately monotonic loss decline; this paper describes **adding isomorphic copies** with error quickly flattening to the irreducible floor. The two shapes differ: one "larger scale better (just slower)," one "after the true independent source count hits bottom." P9's cpu-*proto* measured $b = -4.23 \times 10^{-5}$ (absolute value tiny) precisely $b \approx 0$'s manifestation—15 new homogeneous copies did not significantly pull MSE down from 0.0092.



OASIS (异质个体, 越大越多样) ↔ P9 (同质副本, 加人即饱和) : 差异不在规模, 在个体是否独立

Figure 1 Left: homogeneous expansion— N rises while D unchanged, quality curve flattening by $1/k$ to irreducible floor a ; right: heterogeneous expansion— D rises, irreducible floor a itself pulled down. Horizontal Agent count N . Conceptual schematic, not measured data.

3 External Evidence One: OASIS's "Larger Is More Diverse" Is a Result of Heterogeneous Individuals

3.1 What OASIS Measured

OASIS (Yang et al., arXiv:2411.11581, NeurIPS 2024, CAMEL team) is a large-scale Agent social simulation based on the social media paradigm, supporting up to 1 million user-scale dynamic social networks, diverse action spaces and recommendation systems (paper report). It reproduces information diffusion, group polarization, herding three social phenomena, and observed: **larger Agent group scale leads to stronger group dynamics and more diverse opinions** (paper observation, not independently rechecked).

3.2 Why This Does Not Contradict P9 but Complements It

At first glance this seems to conflict with P9's "homogeneous expansion saturation": OASIS says larger is more diverse, P9 says adding people useless. But putting the two's Agent composition side by side makes it clear—

- OASIS's Agents are **heterogeneous individuals**: different user profiles, different behavior rules, different initial stances, they naturally carry independent information channels (D roughly growing in step with N).
- P9's homogeneous copies are **replications of the same near-correlated source**: same model, same prompt, $\text{corr} \approx 0.9999$, N from 1 to 17 while D always about 2 (strongest single expert + one bad source).

The difference is not "scale," it is "whether individuals are independent." OASIS's "larger is more diverse" holds because its "large" is **heterogeneous individuals' many**; P9's "adding people useless" holds because its "many" is **the same judgment repeated**. Drawing the two curves on one figure (Figure 1), the conclusion is clear: scale $\$N\$$ is not the quality variable, inter-individual information independence is.

3.3 OASIS Boundaries Honestly Pointed Out

OASIS's evidence nature is **phenomenon reproduction**—it can reproduce known social phenomena like polarization, herding, but this does not equal proving the causal mechanism "scale causes polarization" (P27 will specifically discuss social simulation causal validity). This paper only borrows one observation: under the heterogeneous individual premise, scale and diversity grow in step; this with P9's saturation curve under the homogeneous copy premise jointly brackets $\$D\$$'s action interval.

4 External Evidence Two: TUMIX's Heterogeneous Tool Strategies Are an Operable Quality Lever

4.1 What TUMIX Did

TUMIX (Chen et al., arXiv:2510.01279) is a test-time scaling ensemble framework: running multiple Agents in parallel, each adopting **different tool strategies**, then iteratively sharing refinement. The paper reports on Gemini-2.5-Pro/Flash, its average accuracy over the best baseline **up to +3.55%**, with inference cost approximately flat (paper report, not independently rechecked).

4.2 Why This Is "Heterogenization" Not "Adding People"

TUMIX's key is not "parallel multi-Agents," but "each using different tool strategies." If it only ran the same prompt $\$k\$$ times in parallel then voted, per P9's saturation law, most of this +3.55% would be eaten by $\$b\backslash\approx 0\$$. It could measure improvable gains precisely because different tool strategies let different Agents reach **different information channels**—some query knowledge bases, some write code to verify, some search the web—these channels mutually independent, $\$D\$$ growing with strategy count.

This lands Section 2's theoretical inference at an engineering-operable position: **heterogenization is not "changing the model name," but "changing information channels."**

4.3 Corroboration with ClawArena-Team

ClawArena-Team (arXiv:2606.31174, detailed in this volume P17) reports a directionally consistent finding: in sub-Agent orchestration, API cost and management quality approximately decouple, the cheapest configuration instead lands on the Pareto frontier (paper report direction, exact thresholds not independently rechecked). This from the "management ability" dimension corroborates: piling more expensive homogeneous models cannot buy quality, the real lever is structure and information source diversity.

5 Engineering Corollary: Three Questions Before Expanding

Putting Sections 2–4 together, UDOS's Agent legions before expanding should answer three questions in order, not directly spin up new instances.

Table 1 The three "information independence" questions before expanding

Question	If answer "no"	Engineering meaning
Does the new Agent see information existing Agents don't?	$\$D\$$ not increased	adding it only increases $\$N\$$ and the bill, quality pinned at $\$a\$$
Is the new Agent's error near-correlated with existing Agents ($\rho \approx 1$)?	$b \rightarrow 0$	brings no variance decline, P9 saturation curve applies
Is the new Agent's quality verified on an independent calibration set?	weight not calibratable	equal-weight inclusion drags results (P9: 0.2131 vs 0.0092, about $23\times$)

All three questions "yes," expanding buys the diversity dividend; any one "no," expanding buys only the scale narrative.

This directly connects with P9's weighting mechanism: UDOS does not use "one person one vote," but calibrated weighting $w_i = \frac{\exp(-\text{calib_mse}_i / \tau)}{\sum_j \exp(-\text{calib_mse}_j / \tau)}$ ($\tau = 0.05$), allocating voice by historical accuracy on independent calibration sets. This mechanism itself is a structural defense against "homogeneous expansion"—bad sources automatically pressed down in weights, not pouring bias into results like equal weighting.

6 Contrast with Warp-Cortex: Memory Expansion $\neq$ Information Expansion

6.1 The Cost of Singleton Weight Sharing

Warp-Cortex (arXiv:2601.01298, detailed in this volume P22) uses Singleton Weight Sharing (all Agents sharing the same model instance's weights) to press memory complexity from $\mathcal{O}(N \cdot L)$ to $\mathcal{O}(1) + \mathcal{O}(N \cdot k)$, a single RTX 4090 measured about 100 concurrent Agents, 2.2 GB VRAM (paper report). This is a beautiful **memory wall** breakthrough.

But from this paper's view, Singleton Weight Sharing is a **forced homogenization**: all Agents share the same weights, only KV-cache differs. This means they are architecturally designed to "see the same parameters," naturally carrying no independent information. Its value is **lowering the memory cost of enlarging $\$N\$$** , not **building the quality gain of enlarging $\$D\$$** .

6.2 The Memory Wall and Coordination Wall Are Two Walls

Hence this paper and P22's division of labor: Warp-Cortex solves "can memory fit 1000 Agents," this paper answers "after fitting will quality rise." The answer: **memory can expand $\neq$ ability can expand $\neq$ coordination can expand**. Singleton weight sharing dismantled the first wall (memory), but kept the second wall (quality saturation) as-is—because its Agents are homogeneous copies from the start.

This is not criticizing Warp-Cortex, but bounding its value: it is scale infrastructure, not a quality improvement means. Misreading "can run 1000 Agents" as "1000 Agents stronger" is precisely the scale narrative conflation this paper corrects.

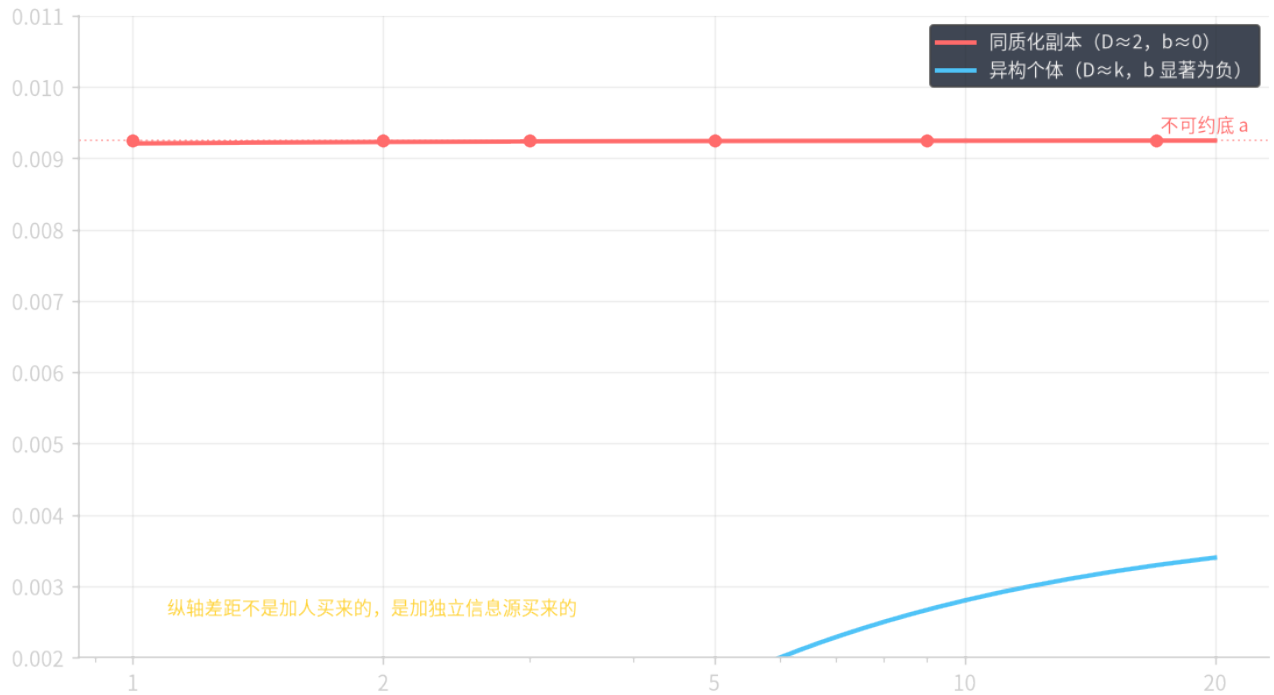


Figure 2 Homogeneous copies (red line): N rises, D unchanged, quality flattening along $1/k$ to irreducible floor a ; heterogeneous individuals (blue line): D rises, a itself pulled down. The two's vertical gap not bought by "adding people," but by "adding independent information sources." Conceptual schematic, not measured data.

7 Cross-Verification with the UDOS Paper Volume

Table 2 This paper's propositions $\leftrightarrow$ paper volume and external evidence comparison

This paper's proposition	Evidence	Evidence grade
quality curve saturating by $a+b/k$	P9 measured $a=0.009252$, $b=-4.23 \times 10^{-5}$	cpu-proto ($k \leq 2$ verified)
homogeneous copies $b \rightarrow 0$, quality flat	P9 after $k > 2$ MSE fluctuates around a	cpu-proto
equal-weight bad source drags about $23\times$	P9: 0.2131 vs 0.0092	verified (two-source comparison)
heterogeneous individuals scale and diversity in step	OASIS observation	paper report • not independently rechecked
heterogeneous tool strategies buy quality	TUMIX +3.55%	paper report • not independently rechecked
cost and management quality decouple	ClawArena-Team	paper report direction • not rechecked
bias-variance-ambiguity decomposition theoretical ancestor	Krogh & Vedelsby 1995	classic literature

8 Dross, Failure Boundaries and Falsifiable Conditions

This paper's arguments have clear failure boundaries, bounded item by item, no synthesis piling.

8.1 Dross and easily misread points

1. **"Larger is more diverse" is not a universal law.** OASIS's observation only holds under the heterogeneous individual premise; if OASIS's Agents were replaced with homogeneous copies (same profile, same behavior rules), "larger is more diverse" would immediately degenerate into P9's saturation curve. Equating scale with diversity is the core conflation this paper opposes.
2. **TUMIX's +3.55% is not evidence of "parallel is universal."** Its gain comes from independent channels brought by heterogeneous tool strategies, not parallelism itself; reading it as "run a few more times and vote to gain points" would again fall into the ≈ 0 trap.
3. **$\$D\$$ is not larger-is-better.** Introducing independent information sources also introduces coordination cost, conflict adjudication cost and inconsistency risk; too large $\$D\$$ may make aggregation itself the bottleneck (P1's fan-in explosion). This paper only claims " $\$D\$$ is the quality variable," not "larger $\$D\$$ always better."

8.2 Failure boundaries

- This paper's saturation law $\$mse(k)=a+b/k\$$ assumes member errors zero-mean, pairwise independent. If the task is **structured geometric solving** (e.g. MAPF path planning, see P21), inter-Agent "errors" are no longer random noise but planning bias, the $\$1/k\$$ form may not hold—that task's coordination success does not rely on error averaging but on global constraint satisfaction.
- This paper does not discuss **causal inference** tasks (needing to identify effects, control confounds), that is P3 RCT and P27 social simulation validity's domain.

8.3 Falsifiable conditions

- **C1:** if in a controlled comparison, homogeneous copies ($corr \approx 0.9999$)'s MSE still significantly declines with $\$k\$$ (significantly breaking below $\$a\$$ and approaching the independent member curve), then P9's "homogeneous saturation" and this paper's $\$b \rightarrow 0\$$ inference are falsified.
- **C2:** if heterogeneous tool strategies (TUMIX-style) under fixed cost repeatedly fail to measure above-noise gains, while pure homogeneous voting instead stably improves, then "independent information sources are the quality variable" must be downgraded to "holds on some tasks."
- **C3:** if a future large-scale system appears whose Agents are all homogeneous (same model, same context) yet quality monotonically rises with $\$N\$$, unexplainable by "a stronger strongest member," then this paper's whole $\$N\$/\$D\$$ dichotomy must be reconstructed.

9 Hard Predictions (2026–2031, post-hoc scorable)

- **F1:** multi-Agent systems published over the next five years, all reporting "continuous linear gains with $\$N\$$ " unable to explain inter-member information independence, their gains in third-party reproductions mostly narrowed to noise or attributed to the strongest single member. (Criterion: in reproduction studies whether the homogeneous copy arm's $\$b\$$ remains significantly negative.)
 - **F2:** at least one widely reproduced "heterogeneous information sources > more homogeneous copies" comparison will appear—i.e. under fixed total compute, 3–5 truly independent information sources (different models/different tools/different data)' ensemble stably beats a legion of 20+ homogeneous copies.
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10 Conclusion

Placing three evidence lines side by side: P9's cpu-proto curve says "homogeneous copies saturate no matter how many added," OASIS's million-scale heterogeneous simulation says "heterogeneous individuals larger is more diverse," TUMIX's heterogeneous tool ensemble says "changing information channels buys measurable gains." The three evidence lines' common conclusion: **the first-principles variable of multi-Agent quality is independent information source count \$D\$, not total Agent count \$N\$.**

For engineering, this proposition gives a cold priority: before spinning up the 1001st Agent, first ask whether it sees a world the first 1000 Agents don't; if the answer is no, it is just one more bill. For research, it gives a killable research program (C1–C3): the homogeneous saturation law either stands in controlled comparison or is overturned by counterexamples. For the whole v7.6.0 volume, it bounds P22 (memory wall) and P30 (topology selection)—architecture can enlarge \$N\$, but only information source heterogenization can raise quality.

Scale buys the accounting office, diversity buys quality. In multi-Agent systems, conflating these two is the most expensive confusion.

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Classic literature

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2026 frontier materials (existence verified 2026-09-21 per EVIDENCE_LEDGER_v76.md; numbers all paper reports, not independently rechecked)

1. Yang, K., et al. (2024). OASIS: Open Agent Social Interaction Simulations with One Million Agents. arXiv:2411.11581, NeurIPS 2024.
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UDOS paper volume (this series, evidence grades marked in text)

1. UDOS v7.5.0 paper volume: P9 "The Quality Saturation Law of Agent Legions and the Failure Boundary of Homogeneous Headcount Expansion," P1 "Million-Scale Extrapolation of Layered Hybrid Topology Coordination Complexity," P0b "Structure Plus Scale, Everything Can Be Scaling Laws," P15 "Life Originates in Structural Organization."
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Evidence Discipline and Reproduction Notes

- This paper is a theoretical/synthesis paper, no new experiments; all UDOS numbers traced to P9 and `reports7/agent_scaling.json`, evidence grades per P9's labels ($\leq$ verified, $>$ cpu-proto, throughput curves constrained by this machine's 2 threads), single-seed results all pilot point estimates, before submission needing ≥ 30 seeds and 95% CI.
- External literature existence, titles and arXiv numbers verified 2026-09-21 per EVIDENCE_LEDGER_v76.md; OASIS's "larger is more diverse" and TUMIX's +3.55% uniformly kept "paper report, not independently rechecked" caliber, not serving as independent evidence for this paper's argument, only as corroboration that "industry phenomena align with UDOS inferences."
- This paper's all predictions (F1–F2) and falsification conditions (C1–C3) post-hoc scorable with public evidence during 2026–2031.